

6	(a) diffraction is the spreading of a wave as it passes through a slit or past an edge	B1	
	when two (or more) waves superpose/meet/overlap	M1	
	resultant displacement is the sum of the displacement of each wave	A1	[3]
	(b) $n\lambda = d \sin \theta$ and $v = f\lambda$	C1	
	max order number for $\theta = 90^\circ$ hence $n (= f/vN) = 7.06 \times 10^{14} / (3 \times 10^8 \times 650 \times 10^3)$	M1	
	$n = 3.6$ hence number of orders = 3	A1	[3]
	(c) greater wavelength so fewer orders seen	A1	[1]